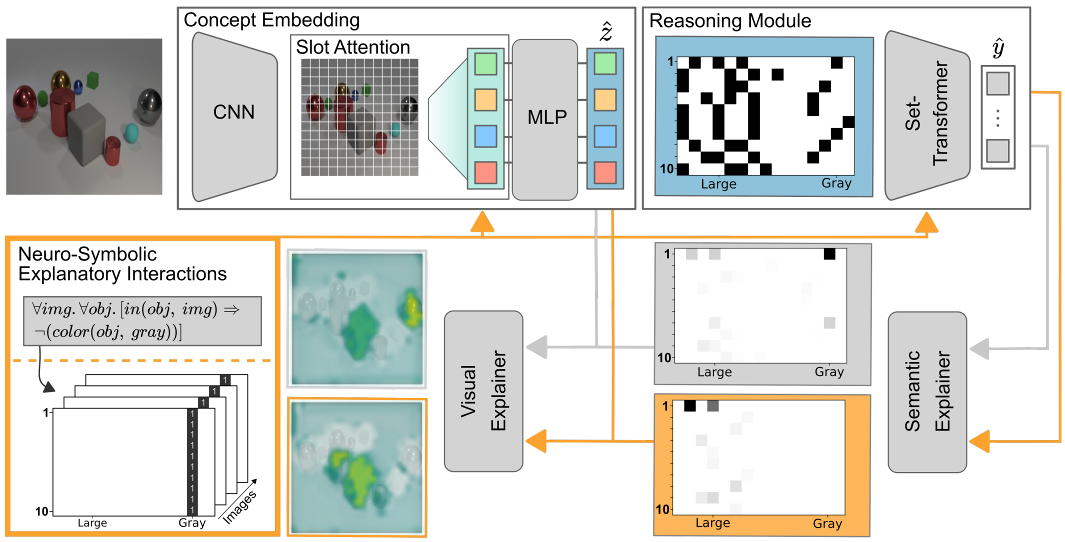

Figure 3: **Neuro-Symbolic XIL for improved explanations and interaction.** (Top) Neuro-Symbolic Concept Learner with Slot-Attention and Set Transformer. (Bottom) Neuro-Symbolic revision pipeline with explanations of the model before (gray) and after applying the feedback (orange).

masks, which we denote as A_i^v.

The semantic user feedback can be in the form of relational functions, φ, for instance, *"if an image belongs to class* 1 *then one object is a large cube"*:

$$\boxed{\forall img. \;\; isclass(img, 1) \;\; \Rightarrow \;\; \exists obj.[in(obj, img) \;\; \wedge \;\; size(obj, large) \wedge shape(obj, cube)] \,,}$$

We define $A_i^s := \bigvee_\varphi A_i^\varphi(\hat{z}_i \models \varphi)$ which describes the disjunction over all relational feedback functions which hold for the symbolic representation, $\hat{z}_i$, of an image, x_i.

An important characteristic of the semantic user feedback is that it can describe different levels of generalizability, so that feedback based on a single sample can be transferred to a set of multiple samples. For instance φ can hold for an individual sample, all samples of a specific class, j, or all samples of the data set. Consequently, the disjunction, $\bigvee_\varphi$, can be separated as: $A^s_{i|y_i=j} = A_i^{sample} \vee A_{c=j}^{class} \vee A^{all}$.

For the sake of simplicity, we are not formally introducing relational logic and consider the semantic feedback in tabular form (*cf.* Fig. 3). To summarize, we have the binary masks for the visual feedback $A_i^v \in [0,1]^M$ and the semantic feedback $A_i^s \in [0,1]^D$.

For the final interaction we refer to XIL with differentiable models and explanation functions, generally formulated as the explanatory loss term, $L_{expl} =$

$$\lambda \sum\nolimits_{i=1}^{N} r(A_i^v,\, \hat{e}_i^h\,) + (1-\lambda) \sum\nolimits_{i=1}^{N} r(A_i^s,\, \hat{e}_i^g\,)\,. \quad (1)$$

Depending on the task, the regularization function, $r(\cdot,\cdot)$, can be the *RRR* term of Ross *et al.* [43] or the *HINT* term of Selvaraju *et al.* [47] (*cf.* Appendix for details on these loss functions). The parameter λ controls how much the different feedback forms are taken into account. Finally, the explanatory loss is concatenated to the original task dependent loss term, *e.g.* the cross-entropy for a classification task.

Reasoning Module. As the output of our concept embedding module represents an unordered set, whose class membership is unaltered by the order of the objects within the set, we require our reasoning module to handle such an input structure. The Set Transformer, recently proposed by Lee *et al.* [24], is a natural choice for such a task.

To generate the explanations of the Set Transformer given the symbolic representation, $\hat{z}_i \in [0,\, 1]^D$, we make use of the gradient-based Integrated Gradients explanation method of Sundararajan *et al.* [49]. Given a function $g: \mathbb{R}^{N\times D} \rightarrow [0,1]^{N\times C}$ the Integrated Gradients method estimates the importance of the jth element from an input sample z_i, z_{ij}, for a model's prediction by integrating the gradients of $g(\cdot)$ along a straight line path from z_{ij} to the jth element of a baseline input, $\tilde{z} \in \mathbb{R}^D$, as $IG_j(z_i) :=$

$$(z_{ij} - \tilde{z}_j) \times \int_{\alpha=0}^{1} \frac{\delta\, g(\tilde{z} + \alpha \times (z_i - \tilde{z}))}{\delta z_{ij}} \delta\alpha\,. \quad (2)$$

Given the input to the Set Transformer, $\hat{z} \in [0,\, 1]^{N\times D}$, and $\tilde{z} = \mathbf{0}$ as a baseline input, we finally apply a zero threshold to only receive positive importance and thus have:

$$\hat{e}_i^g := \sum\nolimits_{j=1}^{D} min(IG_j(\hat{z}_i),\, 0)\,. \quad (3)$$

(Slot) Attention is All You Need (for object-based explanations). Previous work of Yi *et al.* [57] and Mao *et*

Data Set	Size	Classes	Input-dimensions	Multi-object	Visual confounder	Non-visual confounder	Number of rule-types
ToyColor [43]	40k	2	$5 \times 5 \times 3$	✗	✓	✗	1
ColorMNIST [17]	70k	10	$28 \times 28 \times 3$	✗	✗	✓	1
Decoy-MNIST [43]	70k	10	$28 \times 28 \times 3$	✗	✓	✗	1
Plant Data Set [44]	2.4k	2	$213 \times 213 \times 64$	✗	✓	✗	1
ISIC Skin Cancer Data Set [4, 51]	21k	2	$650 \times 450 \times 3$	✗	✓	✗	1
Our CLEVR-Hans3	13.5k	3	$320 \times 480 \times 3$	✓	✓	✓	2
Our CLEVR-Hans7	31.5k	7	$320 \times 480 \times 3$	✓	✓	✓	4

Table 1: **The complexity of CLEVR-Hans.** The CLEVR-Hans data sets represent confounded data sets in which the confounding factors are not separable in the original input space. Additionally, more than one conceptual rule must be applied in order to revise the model.

al. [31] has shown an interesting approach for creating a Neuro-Symbolic concept leaner based on a Mask-RCNN [11] scene parser. For our concept learner, we make use of the recent work of Locatello *et al.* [28]. Their proposed Slot Attention module allows to decompose the hidden representation of an encoding space into a set of task-dependent output vectors, called "slots". For example, the image encoding of a CNN backbone can be decomposed such that the hidden representation is separated into individual slots for each object. These decomposed slot encodings can then be used in further task-dependent steps, *e.g.* attribute prediction of each object. Thus with Slot Attention, it is possible to create a fully differentiable object-centric representation of an entire image without the need to process each object of the scene individually in contrast to the system of [57, 31].

An additional important feature of the Slot Attention module for our setting is the ability to map each slot to the original input space via the attention maps. These attention maps are thus natural, intrinsic visual explanations of the detected objects. In contrast, with the scene parser of [57, 31] it is not as straightforward to generate visual explanations based on the explanations of the reasoning module. Consequently, using the Slot Attention module, we can formulate the dot-product attention for a sample x_i, as

$$B_i := \sigma\left(\frac{1}{\sqrt{D'}}\, k(F_i) \cdot q(S_i)^T\right) \in \mathbb{R}^{P \times K}\,, \qquad (4)$$

where σ is the softmax function over the slots dimension, $k(F_i) \in \mathbb{R}^{P \times D'}$ a linear projection of the feature maps F_i of an image encoder for x_i, $q(S_i) \in \mathbb{R}^{K \times D'}$ a linear projection of the slot encodings S_i and $\sqrt{D'}$ a fixed softmax temperature. P represents the feature map dimensions, K the number of slots and D' the dimension which the key and query functions map to.

Finally, we can formulate $E^h(h(\cdot),\ \hat{e}_i^g)$ based on the attention maps B_i, and the symbolic explanation $\hat{e}_i^h$. Specifically, we only want an explanation for objects which were identified by the reasoning module as being relevant for the final prediction:

$$\hat{e}_i^h := \sum_{k=1}^{K} \begin{cases} B_{ik}, & \text{if } max(\hat{e}_{ik}^g) \geq t \\ \mathbf{0} \in \mathbb{R}^P, & \text{otherwise} \end{cases}\,, \qquad (5)$$

where t is a pre-defined importance threshold. Alternatively the user can manually select explanations for each object.

Interchangeability of the Modules. Though both Slot-Attention and Set Transformer have strong advantages as stated above, alternatives exist. Deep Set Prediction Networks [58], Transformer Set Prediction Networks [20] or Mask-RCNN based models [11] are viable alternatives to the Slot Attention module as concept embedding module. The generation of visual explanations within these models, *e.g.* via gradient-based explanation methods, however, is not as straightforward. Truly rule-based classifiers [38, 29], logic circuits [26], or probabilistic approaches [5, 37, 19, 30], are principally viable alternatives for the Set Transformer, though it remains preferable for this module to handle unordered sets.

5. The CLEVR-Hans Data Set

Several confounded computer vision data sets with varying properties, *e.g.* number of classes, already exist. Tab. 1 provides a brief overview of such data sets. We distinguish here between the number of samples, number of classes, image dimensions, and whether an image contains multiple objects. More important are whether a confounding factor is spatially separable from the relevant features, *e.g.* the colored corner spots in Decoy-MNIST, whether the confounding factor is not visually separable, *e.g.* the color in ColorMNIST that superimposes the actual digits, and finally, once the confounding factor has been identified, how many different conceptual rule-types must be applied in order to revise the model, *i.e.* the corner rule for the digits in Decoy-MNIST is the same, regardless of which specific class is being considered.

To the best of our knowledge, the confounded data sets listed in Tab.1, apart from ColorMNIST, possess spatially separable confounders. One can, therefore, revise a model by updating its spatial focus. However, this is not possible if the confounding and true factors are not so easily separable in the input dimensions.

The CLEVR data set of [14] is a particularly interesting data set, as it was originally designed to diagnose reasoning modules and presents complex scenes consisting of mul-

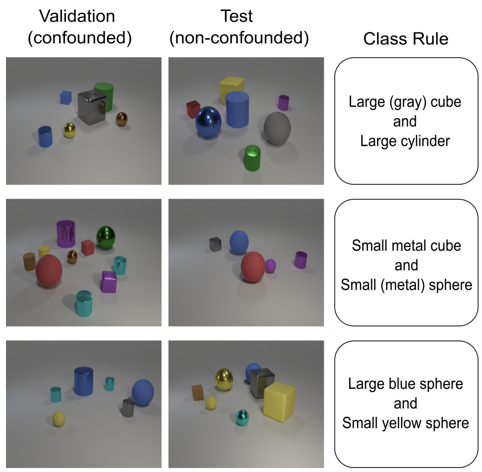

Figure 4: **Schematic of the CLEVR-Hans3 data set.** Attributes in brackets are the confounding factors in the train and validation sets.

tiple objects and different relationships between these objects. Using the available framework of [14], we have thus created a new confounded data set, which we refer to as the CLEVR-Hans data set. This data set consists of CLEVR images divided into several classes. The membership of a class is based on combinations of objects' attributes and relations. Additionally, certain classes within the data set are confounded. Thus, within the data set, consisting of train, validation, and test splits, all train, and validation images of confounded classes will be confounded with a specific attribute or combination.

We have created two variants of this data set[2], which we refer to as CLEVR-Hans3 and CLEVR-Hans7. CLEVR-Hans3 contains three classes, of which two are confounded. Fig. 4 shows a schematic representation of this data set. Images of the first class contain a large cube and large cylinder. The large cube has the color gray in every image of the train and validation set. Within the test set, the color of the large cube is shuffled randomly. Images of the second class contain a small sphere and small metal cube. The small sphere is made of metal in all training and validation set images, however, can be made of either rubber or metal in the test set. Images of the third class contain a large blue sphere and a small yellow sphere in all images of the data set. This class is not confounded. CLEVR-Hans7 contains seven classes, of which four are confounded. This data set, next to containing more class rules, also contains more complex class rules than CLEVR-Hans3, *e.g.* class rules are also based on object positions. Each class in both data sets consists of 3000 training, 750 validation, and 750 test images.

Finally, the images were created such that the exact combinations of the class rules did not occur in images of other classes. It is possible that a subset of objects from one class rule occur in an image of another class. However, it is not possible that more than one complete class rule is contained in an image. In summary, these data sets present an opportunity to investigate confounders and model decisions for complex classification rules within a benchmark data set that is more complex than previously established confounded data sets (see Tab. 1).

[2] `https://github.com/ml-research/CLEVR-Hans`

6. Experimental Evidence

Our intention here is to investigate the benefits of Neuro-Symbolic Explanatory Interactive Learning. To this end, we make use of our CLEVR-Hans data sets to investigate (1) the downsides of deep learning (DL) models in combination with current (visual) XAI methods and, in comparison, (2) the advantages of our NeSy XIL approach. In particular, we intend to investigate the benefits of neuro-symbolic explanations to not just provide more detailed insights of the learned concept, but allow for better interaction between human users and the model's explanations. We present qualitative as well as quantitative results for each experiment. *Cf.* Appendix for further details on the experiments and implementation, and additional qualitative results.

Architectures. We compared our Neuro-Symbolic architecture to a ResNet-based CNN model [12], which we denote as CNN. For creating explanations of the CNN, we used the Grad-CAM method of Selvaraju *et al.* [46], a backpropagation based explanation method that visualizes the gradients of the last hidden layer of the network's encoder, and represents a trade-off between high visual representation and spatial information.

Due to the modular structure of our Neuro-Symbolic concept learner, Clever-Hans behavior can be due to errors within its sub-modules. As previous work [43, 47, 50, 44] has already shown how to revise visual explanations, we did not focus on revising the visual explanations of the concept learner for our experiments. Instead, we assumed the concept embedding module to produce near-perfect predictions and visual explanations and focused on revising the higher-level explanations of the reasoning module. Therefore, we employed a Slot-Attention module pre-trained supervisedly on the original CLEVR data set [28].

Preprocessing. We used the same pre-processing steps as the authors of the Slot-Attention module [28].

Training Settings. We trained the two models using two settings: A standard classification setting using the cross-entropy loss (Default) and the XIL setting where the explanatory loss term (Eq. 1) was appended to the cross-entropy term. The exact loss terms used will be discussed in the corresponding subsections.

User Feedback. As in [50, 47, 44], we simulated the user feedback. The exact details for each experiment can be found in the corresponding subsections.